BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 6 — Stoichiometry

Total Marks: 46	**Time Allowed: 1h 45m**	**Version: 1**

General Instructions:

1. Write your name, roll number and section clearly on the answer sheet.
2. Section A is compulsory. Encircle the correct option on the question paper itself.
3. In Section B, attempt any **8** questions out of 11. For each question, attempt either the main question **OR** its alternative.
4. In Section C, attempt any **2** questions out of 4. For each question, attempt either the main question **OR** its alternative.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The empirical formula of glucose ($C_6H_{12}O_6$) is:	$C_6H_{12}O_6$	CH_2O	$C_2H_4O_2$	CHO	ⒶⒷⒸⒹ
2	The molecular mass of naphthalene ($C_{10}H_8$) is:	120 amu	108 amu	128 amu	136 amu	ⒶⒷⒸⒹ
3	The formula mass of $Mg(OH)_2$ is: (Mg=24, O=16, H=1)	40 amu	56 amu	58 amu	74 amu	ⒶⒷⒸⒹ
4	One mole of any substance contains how many representative particles?	6.022×10^{22}	6.022×10^{23}	6.022×10^{24}	6.022×10^{20}	ⒶⒷⒸⒹ
5	How many moles of molecules are there in 16 g of oxygen gas (O_2)?	1 mol	0.5 mol	0.1 mol	2 mol	ⒶⒷⒸⒹ
6	The mass of 9.05 moles of ozone (O_3) is: (O=16)	48 g	144 g	434.4 g	480 g	ⒶⒷⒸⒹ
7	How many atoms are present in 1.25 moles of zinc (Zn)?	6.022×10^{23}	7.53×10^{23}	4.82×10^{23}	1.25×10^{23}	ⒶⒷⒸⒹ
8	The formula of aluminium oxide (Al^{3+}, O^{2-}) is:	AlO	Al_2O_3	Al_3O_2	AlO_3	ⒶⒷⒸⒹ
9	Which compound has the same empirical and molecular formula?	H_2O_2	C_6H_6	H_2O	$C_6H_{12}O_6$	ⒶⒷⒸⒹ
10	What is the mass of 4 moles of hydrogen gas H_2? (H=1.008)	8.064 g	4.032 g	1 g	1.008 g	ⒶⒷⒸⒹ
11	In a chemical equation, balancing must be done by changing:	subscripts	symbols	coefficients	state symbols	ⒶⒷⒸⒹ
12	In an ionic equation, ions that appear on both sides and do not participate are called:	reactive ions	spectator ions	anions	cations	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt any 8 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	Define empirical formula and molecular formula. Give one example of each.	**3**	**OR**	What is a mole? State Avogadro's number and name three types of representative particles.	**3**
2(ii)	Calculate the molecular mass of glucose $C_6H_{12}O_6$. (C=12, H=1.008, O=16)	**3**	**OR**	Calculate the formula mass of $CuSO_4\cdot5H_2O$. (Cu=63.5, S=32, O=16, H=1)	**3**
	Differentiate between gram atomic			Differentiate between molecular	

2(iii)	mass, gram molecular mass, and gram formula mass.	**3**	**OR**	formula and empirical formula. Which two quantities are needed to find the molecular formula from the empirical formula?	**3**
2(iv)	Calculate the mass of 0.25 moles of CO_2. (C=12, O=16)	**3**	**OR**	A balloon is filled with 5 g of hydrogen gas (H_2). Calculate the number of moles of H_2. (H=1.008)	**3**
2(v)	How many molecules are present in 0.5 moles of methane (CH_4)?	**3**	**OR**	A sample of titanium contains 3.011×10^{23} Ti atoms. Calculate the number of moles of Ti. (N_A = 6.022×10^{23})	**3**
2(vi)	Write the chemical formula of: (a) sodium chloride (Na^+, Cl^-) (b) magnesium nitride (Mg^{2+}, N^{3-}) (c) potassium sulphate (K^+, SO_4^{2-})	**3**	**OR**	Write the structural formula and molecular formula of n-butane (C_4H_{10}). Identify the structural formula of CH_3-CH_2-OH and write its molecular formula.	**3**
2(vii)	Balance the following chemical equation: $CH_{4(g)} + O_{2(g)} \rightarrow CO_{2(g)} + H_2O_{(l)}$	**3**	**OR**	Balance the following chemical equation and add state symbols: $Na_{(s)} + H_2O_{(l)} \rightarrow NaOH_{(aq)} + H_{2(g)}$	**3**
2(viii)	How many moles of molecules are in 3.011×10^{22} molecules of formaldehyde (CH_2O)? Also find its mass. (C=12, H=1, O=16)	**3**	**OR**	Calculate the number of atoms in 0.2 moles of aluminium (Al). (N_A = 6.022×10^{23})	**3**
2(ix)	Write the steps for writing the chemical formula of a binary ionic compound. Apply the steps to write the formula of aluminium oxide (Al^{3+}, O^{2-}).	**3**	**OR**	Define ionic equation. Write the ionic equation for the reaction: $HCl_{(aq)}$ + $NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$	**3**
2(x)	Calculate the number of moles in: (a) 15 g of NaCl (b) 40 g of sulphur (S). (Na=23, Cl=35.5, S=32)	**3**	**OR**	Calculate the mass of: (a) 1.2 moles of potassium (K) (b) 0.25 moles of steam (H_2O). (K=39, H=1.008, O=16)	**3**
2(xi)	Aspirin ($C_9H_8O_4$) is used as a mild painkiller. (a) Write its empirical formula. (b) Calculate its molecular mass. (C=12, H=1, O=16)	**3**	**OR**	Caffeine is $C_8H_{10}N_4O_2$. (a) Write its empirical formula. (b) Calculate its molecular mass. (C=12, H=1, N=14, O=16)	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt any 2

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Define molecular mass and formula mass. Give one example of each with full calculation. [2] (b) Calculate the formula mass of $KClO_3$ (potassium chlorate) and $NaHCO_3$ (baking soda). (K=39, Cl=35.5, O=16, Na=23, H=1, C=12) [3]	**5**	**OR**	(a) Define empirical formula and molecular formula. Explain with benzene (C_6H_6) and hydrogen peroxide (H_2O_2) as examples. [2] (b) Write empirical formulas for: (i) Aspirin $C_9H_8O_4$ (ii) Vinegar CH_3COOH (iii) TNT $C_7H_5N_3O_6$. Also calculate the molecular mass of each. [3]	**5**
3(ii)	(a) Define mole and Avogadro's number. Explain the difference between gram atomic mass, gram molecular mass and gram formula mass. [3] (b) Calculate the number of molecules in: (i) 2.5 moles of CO_2 (ii) 3.4 moles of NH_3. [2]	**5**	**OR**	(a) Using the mole concept, calculate the number of moles in: (i) 2.4 g of He (ii) 250 mg of carbon (iii) 40 g of sulphur. (He=4, C=12, S=32) [3] (b) Calculate the mass of: (i) 75 moles of H_2 (ii) 0.15 moles of H_2SO_4. (H=1.008, S=32, O=16) [2]	**5**

3(iii)	(a) Describe the steps for balancing a chemical equation. Why is it important to balance chemical equations? [2] (b) Balance the following equations and add correct state symbols: (i) $NH_3 \rightarrow N_2 + H_2$ (ii) $Al + HCl \rightarrow AlCl_3 + H_2$ (iii) $Fe + O_2 \rightarrow Fe_2O_3$ [3]	**5**	**OR**	(a) Define ionic equation. Explain the concept of spectator ions and state when they are removed. [2] (b) Transform the following into ionic equations: (i) $Mg_{(s)} + H_2SO_{4(aq)} \rightarrow MgSO_{4(aq)} + H_{2(g)}$ (ii) $AgNO_{3(aq)} + NaCl_{(aq)} \rightarrow AgCl_{(s)} + NaNO_{3(aq)}$ (iii) $Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}$ [3]	**5**
3(iv)	(a) A 0.5 g sample of potassium (K) was added to water. [K=39, N_A=6.022×10^{23}] $K_{(s)} + H_2O_{(l)} \rightarrow KOH_{(aq)} + H_{2(g)}$ (i) Balance the equation. [1] (ii) Calculate the number of moles of K. [2] (iii) Calculate the number of K atoms in the sample. [2]	**5**	**OR**	(a) Potassium chlorate ($KClO_3$) decomposes on heating to produce KCl and O_2. [K=39, Cl=35.5, O=16] (i) Write and balance the equation for this decomposition. [1] (ii) Calculate the formula mass of $KClO_3$. [2] (iii) How many moles of $KClO_3$ are in 24.5 g? [2]	**5**

Invigilator's Signature: ____________________ Candidate's Signature: ____________________

Chapters Covered: Chapter 6 — Stoichiometry (pp. 95–112)

— End of Question Paper —